

Notes: Greenwood (RFS, 2008)

Excess Comovement of Stock Returns: Evidence from Cross-Sectional Variation in Nikkei 225 Weights

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1 Big picture

- **Question.** Does correlated demand from index traders create *excess comovement* in stock returns, and does the amount of excess comovement vary systematically across stocks with different index weights?
- **Setting.** The paper studies Japan's **Nikkei 225**, a price-weighted stock index whose methodology creates large, persistent, and partly arbitrary cross-sectional differences in constituent weights. The sample covers **1993–2003**.
- **Core empirical challenge.** Event studies of index additions and deletions show that membership affects return comovement, but those designs are vulnerable to endogeneity: stocks that enter an index may already differ in size, liquidity, visibility, or information environment. The paper therefore wants a cleaner source of variation *within* index members.
- **Main conceptual contribution.** Instead of asking whether comovement changes when a stock enters an index, the paper asks whether stocks that are more heavily *overweighted* inside the same index comove more with the index and less with nonindex stocks. This cross-sectional design is the paper's main innovation.
- **Main hypotheses.**
 - **Hypothesis 1:** within the Nikkei 225, a stock's covariance with other index stocks should rise with its index overweighting.
 - **Hypothesis 2:** a stock's covariance with nonindex stocks should fall with its index overweighting.
 - **Hypothesis 3:** if contemporaneous comovement is excessive rather than fundamental, overweighted stocks should later *reverse* more strongly; their returns should load more negatively on *lagged* index returns.
- **Main empirical design.** The author estimates repeated 100-day first-stage time-series regressions to measure each stock's comovement with equal-weighted Nikkei returns and with the broader TOPIX market return. He then runs Fama-MacBeth style cross-sectional regressions of these comovement measures on lagged index overweighting.
- **Main baseline result.** In the core specification, index overweighting strongly predicts comovement with other Nikkei stocks. In Table 3, the average cross-sectional slope of index beta on overweighting is about **0.218**, and overweighting explains about **21%** of the cross-sectional variation in daily comovement among Nikkei stocks.
- **Key nonindex result.** In conditional regressions that include the TOPIX return, overweighting predicts *less* comovement with nonindex stocks. This sign pattern is exactly what a funds-switching / common-demand story predicts.
- **Key predictability result.** Overweighted stocks not only comove more with the Nikkei today; they also reverse more tomorrow. Stocks above the median overweighting have contemporaneous index beta around **1.18** versus **0.92** for underweighted stocks, but lagged-index beta around **-0.066** versus **+0.05**. A trading strategy that bets on this reversal earns a very high Sharpe ratio, roughly **above 2.3**.

- **Main credibility result.** The main patterns survive controls for size and price-to-book, expanded-sample regressions with an index-membership dummy, longer measurement windows, alternative ways of constructing the index return, leverage controls, and checks against nonsynchronous-trading bias.
- **Main interpretation.** The evidence is most consistent with a **demand-based** view of excess comovement: common trading by index investors generates correlated temporary mispricing, and arbitrage is not strong enough to remove it immediately.
- **Economic takeaway.** Arbitrary index design can influence not only levels of stock prices but also the correlation structure of returns. Cross-sectional differences in index weights matter for how much stocks move together and for how quickly those movements reverse.

2 Data and measurement (institutional context and key objects)

2.1 Why the Nikkei 225 is the right laboratory

- The Nikkei 225 is not market-value weighted. After adjusting prices for face value, it is essentially **price weighted**.
- This means some stocks receive much larger index weights than they would in a value-weighted benchmark such as the TOPIX.
- The paper notes that some Nikkei constituents are overweighted by a factor of **10 or more** relative to their value-weighted benchmark weights.
- The index is also economically important: Nomura estimated that about **2.43 trillion yen** were benchmarked to the Nikkei in April 2000, and Nikkei 225 futures trading volume exceeded TOPIX futures volume.

Interpretation. The Nikkei 225 creates large and persistent cross-sectional variation in exposure to index-trader demand, while remaining a market that investors actually care about.

2.2 Sample construction

- Main sample: **298 stocks** that were members of the Nikkei 225 for at least 200 days between **1 September 1993** and **29 August 2003**.
- At any point in time, about **225** stocks are in the index.
- Of these 298 stocks, **164** remain in the index for the full sample, while **73** are added and **73** are deleted during 1993–2003.
- Expanded sample: an additional **1458** Japanese stocks that are never in the Nikkei during the sample but have at least two years of returns data.

- Stock return, volume, and accounting variables come from **Datastream**; index membership changes are assembled from the *Nihon Keizai Shimbun* webpage.

Interpretation. The paper builds both a within-index panel and a broader all-stocks panel, allowing it to separate the role of index membership from the role of index *weighting*.

2.3 How the Nikkei 225 is constructed

The Nikkei level on day t is:

$$P_{N225,t} = \frac{\sum_{i=1}^{225} \frac{P_{i,t}}{FV_i/50}}{D_t}, \quad (1)$$

where $P_{i,t}$ is stock i 's price, FV_i is the stock's face value, and D_t is the index divisor.

Important implications:

- A stock's index weight depends on its **face-value-adjusted price**, not on its market capitalization.
- The divisor preserves continuity in the index level after splits or capital changes, but not necessarily in constituent weights.
- Most firms have face value **50**, but some have face values **500**, **5000**, or **50,000**, which creates additional dispersion in effective index weights.

Interpretation. This methodology makes the Nikkei 225 an unusually sharp setting in which index weights diverge from economic size.

2.4 Index returns and Nikkei weights

The Nikkei return can be written as:

$$R_{N225,t} = \sum_{i=1}^{225} w_{i,t}^{N225} R_{i,t}, \quad (2)$$

where the constituent weight is

$$w_{i,t}^{N225} = \frac{P_{i,t}/(FV_i/50)}{\sum_{j=1}^{225} P_{j,t}/(FV_j/50)}. \quad (3)$$

- The paper also constructs an **equal-weighted Nikkei return**, denoted R_t^{EWN225} .
- This is used for most empirical tests because it captures the average comovement with index stocks and avoids a mechanical relation between a stock's own index weight and the return measure on the right-hand side.
- The equal-weighted Nikkei return is highly correlated with the actual Nikkei return (about **93%**) over the sample.

Interpretation. Using the equal-weighted index return makes the tests cleaner: if overweighted stocks comove more, it is not just because they mechanically contribute more to the index return itself.

2.5 The main explanatory variable: index overweighting

The paper first compares a stock's Nikkei weight to its value-weighted benchmark weight in the TOPIX:

$$\frac{w_{i,t}^{N225}}{w_{i,t}^{VW}}. \quad (4)$$

It then defines the main overweighting measure as

$$OW_{i,t} = \log \left(1 + \frac{w_{i,t}^{N225}}{w_{i,t}^{VW}} \right). \quad (5)$$

- For nonindex stocks, Nikkei weight is zero, so overweighting is zero.
- The paper reports that the average Nikkei weight exceeds the TOPIX weight by a factor of about **5.7**.
- Overweighting is highly persistent over time because stock returns raise both Nikkei weights and value weights in broadly similar proportion.

Interpretation. Overweighting is the paper's proxy for a stock's exposure to common index-trader demand shocks.

2.6 Why lagged overweighting is used

- In the main tests, overweighting is measured at the **start** of the 100-day return window used to estimate comovement.
- This timing choice prevents the comovement estimate from mechanically feeding back into the explanatory variable.
- In practice, this lag matters little because cross-sectional weight ranks are very stable over time.

Interpretation. The paper wants overweighting to behave like a predetermined exposure variable, not an outcome measured simultaneously with returns.

2.7 Main outcome variables

The paper studies four closely related objects.

1. Index beta from a univariate regression. This is the stock's loading on the equal-weighted Nikkei return.

2. R^2 from the same regression. This captures how tightly the stock's returns are related to the returns of other Nikkei stocks.

3. Conditional beta on the equal-weighted Nikkei return. This comes from a bivariate regression that also controls for the broader TOPIX return.

4. Conditional beta on the TOPIX return. This measures comovement with stocks outside the Nikkei after controlling for index comovement.

Interpretation. The paper alternates between raw comovement, conditional comovement, and reversal of comovement. All three should line up if common demand is the right mechanism.

2.8 Controls and auxiliary variables

The main controls are:

- log market capitalization (**Size**),
- log price-to-book ratio (**P/B**),
- a dummy for Nikkei index membership in the expanded-sample regressions,
- leverage in robustness checks.

Interpretation. These controls are meant to absorb standard cross-sectional differences in risk and return behavior, so that the coefficient on overweighting is not merely picking up well-known size or value effects.

2.9 Institutional facts worth remembering

- Greenwood (2005) shows that additions to the Nikkei 225 experience large abnormal returns, above **10%** on average.
- Average daily Nikkei 225 futures trading volume is estimated at roughly **581 billion yen**, versus about **271 billion yen** for TOPIX futures.
- The paper treats these facts as evidence that the Nikkei is a genuine trading habitat, not merely a statistical index.

Interpretation. The comovement results matter economically because the index is actively tracked and traded.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework and cross-sectional predictions

The paper builds on limits-to-arbitrage models with two types of agents:

- **index traders**, who move funds into and out of a basket of index stocks in fixed proportions;

- **arbitrageurs**, who lean against these flows but cannot fully eliminate the price impact.

If index traders buy stock A ten times as aggressively as stock B , both stocks should rise together when index demand is high, but A should react more strongly. This logic produces three empirical predictions.

1. Stocks with higher overweighting should have higher contemporaneous comovement with index stocks.
2. These same stocks should have lower conditional comovement with nonindex stocks.
3. Because the demand shocks are not purely fundamental, high-overweighting stocks should subsequently reverse more.

Interpretation. The model is not used for structural estimation. It is mainly a disciplined way to generate sign predictions for the cross-section.

3.2 Univariate comovement regressions

For each stock i , the basic 100-day time-series regression is

$$R_{i,t} = \alpha_i + \beta_i^{EWN225} R_t^{EWN225} + \varepsilon_{i,t}. \quad (6)$$

The paper then records:

- β_i^{EWN225} , the stock's comovement with Nikkei stocks;
- R_i^2 , the fraction of its return variation explained by Nikkei returns.

Interpretation. The slope captures covariance with the index. The R^2 captures how much of the stock's overall movement is tied to the index.

3.3 Rolling cross-sectional regressions for daily comovement

Every 100 days, the paper regresses the estimated comovement measures on lagged overweighting:

$$\beta_{i,t}^{EWN225} = a_t + b_t OW_{i,t-1} + u_{i,t}, \quad (7)$$

and

$$R_{i,t}^2 = a_t + b_t OW_{i,t-1} + u_{i,t}. \quad (8)$$

The controlled version is:

$$\beta_{i,t}^{EWN225} = a_t + b_t OW_{i,t-1} + c_t \ln(P/B)_{i,t-1} + d_t Size_{i,t-1} + u_{i,t}, \quad (9)$$

with an analogous equation for $R_{i,t}^2$.

- The paper averages the estimated a_t, b_t, c_t, d_t across cross-sections using Fama-MacBeth style inference.

- There are **26 nonoverlapping 100-day cross-sections** in the main panel, with about **222 stocks** per cross-section on average.

Interpretation. The coefficient b_t is the main object of interest. A positive b_t means stocks more exposed to index trading move more with the index.

3.4 Conditional comovement with Nikkei and nonindex stocks

To separate index comovement from broader market exposure, the paper estimates

$$R_{i,t} = \alpha_i + \beta_{i,t}^{*EWN225} R_t^{EWN225} + \beta_{i,t}^{TOPIX} R_t^{TOPIX} + \varepsilon_{i,t}. \quad (10)$$

It then runs cross-sectional regressions:

$$\beta_{i,t}^{*EWN225} = a_t + b_t OW_{i,t-1} + c_t \ln(P/B)_{i,t-1} + d_t Size_{i,t-1} + u_{i,t}, \quad (11)$$

and

$$\beta_{i,t}^{TOPIX} = a_t + b_t OW_{i,t-1} + c_t \ln(P/B)_{i,t-1} + d_t Size_{i,t-1} + v_{i,t}. \quad (12)$$

Interpretation.

- A positive coefficient in the first equation says overweighted stocks comove more with other Nikkei stocks even after controlling for market exposure.
- A negative coefficient in the second equation says overweighted stocks comove less with nonindex stocks.

The sign reversal between these two equations is one of the paper's most distinctive empirical predictions.

3.5 Expanded-sample design: membership versus weighting

In the larger sample that includes nonindex stocks, the paper adds an index-membership dummy:

$$\beta_{i,t} = a_t + b_t OW_{i,t-1} + c_t 1\{OW_{i,t-1} > 0\} + d_t \ln(P/B)_{i,t-1} + e_t Size_{i,t-1} + u_{i,t}. \quad (13)$$

The same logic is applied to $\beta_{i,t}^{*EWN225}$ and $\beta_{i,t}^{TOPIX}$.

Interpretation. This asks a sharper question: after one controls for whether a stock is in the Nikkei at all, does *extra* overweighting inside the index still matter? The answer is yes.

3.6 Longer-horizon comovement

The paper repeats the comovement exercise using **weekly** and **monthly** returns in place of daily returns:

$$\beta_{i,t,h}^{EWN225} = a_t + b_t OW_{i,t-1} + u_{i,t}, \quad (14)$$

where $h \in \{\text{daily, weekly, monthly}\}$.

Interpretation. If excess comovement is short-lived and later corrected, its cross-sectional signature should be strongest at high frequency and weaker at longer horizons.

3.7 Direct test of reversal

To measure both contemporaneous comovement and next-day reversal, the paper estimates

$$R_{i,t} = \alpha_i + \beta_{i,t}^{EWN225} R_t^{EWN225} + \beta_{i,t-1}^{EWN225} R_{t-1}^{EWN225} + \varepsilon_{i,t}. \quad (15)$$

Then it studies the cross-section of the lagged-beta term:

$$\beta_{i,t-1}^{EWN225} = a_t + b_t OW_{i,t-1} + u_{i,t}. \quad (16)$$

Interpretation. A negative b_t here means the stocks that comove most with the index today are exactly the ones that reverse most tomorrow.

3.8 Trading strategy implied by excess comovement

The paper forms a zero-investment portfolio that shorts overweighted stocks after positive index returns and buys them after negative index returns:

$$w_{i,t} = - \left(OW_{i,t-1} - \frac{1}{N} \sum_{j=1}^N OW_{j,t-1} \right) R_{t-1}^{EWN225}. \quad (17)$$

- If the Nikkei rose yesterday, the strategy shorts stocks with above-average overweighting and goes long stocks with below-average overweighting.
- If the Nikkei fell yesterday, it does the opposite.

Interpretation. This is a direct bet that the demand-driven part of comovement reverses in the cross-section.

3.9 Robustness checks

The paper reports several robustness exercises.

- **Longer measurement window:** use 250 days rather than 100 days to estimate comovement.
- **Alternative controls:** add leverage to the cross-sectional regressions.
- **Alternative return aggregates:** replace equal-weighted Nikkei returns with the actual Nikkei return, the equal-weighted return of current constituents, or the value-weighted return of constituents.
- **Nontrading bias:** split stocks by trading volume and inspect tick data near the close.

Interpretation. The paper tries to show that the main results are not fragile to reasonable changes in measurement or to market-microstructure concerns.

4 Identification and empirical design (what makes the estimates credible)

4.1 Why the cross-sectional design is cleaner than event studies

Event studies of index additions and deletions compare a stock before and after it enters the index. But index entrants may change in many ways at once: they may become larger, more visible, more liquid, or more institutionally important. This paper instead studies *differences across stocks already inside the same trading habitat*.

Interpretation. The design asks whether stocks that index traders buy in larger proportions move more with the rest of the basket. That is a more direct test of a common-demand mechanism.

4.2 Why lagged overweighting is plausibly predetermined

- Overweighting is measured before the 100-day window used to estimate comovement.
- Much of the cross-sectional variation comes from the Nikkei's arbitrary weighting methodology, especially face-value adjustments and the fact that the index is price weighted rather than value weighted.
- The paper argues that the ranking of overweighting is highly stable over time, so short-run return shocks do not drive the results.

Interpretation. The design is not claiming that weights are random in an absolute sense. It is claiming that, conditional on being in the Nikkei, the differences in *relative* weights are a useful and fairly predetermined exposure to common demand.

4.3 How the paper deals with common fundamentals

A natural alternative story is that high-overweighting stocks simply have higher market exposure for fundamental reasons. The paper addresses this in several ways:

- it uses conditional regressions that include the broad TOPIX return;
- it controls for size and price-to-book in the cross-section;
- it shows that the sign flips when the dependent variable is comovement with *nonindex* stocks;
- it studies later reversal, which common fundamentals alone do not naturally imply.

Interpretation. No single control is definitive, but the whole pattern is hard to reconcile with a purely fundamental factor model.

4.4 Why the negative relation with nonindex comovement matters

If overweighted Nikkei stocks just happened to be more market sensitive, one would expect them to comove more with *all* stocks. Instead, the paper finds the opposite for nonindex stocks once index comovement is isolated.

Interpretation. This is exactly what one expects if money is flowing into the Nikkei basket and away from other stocks, rather than if overweighted stocks simply carry more systematic risk.

4.5 Why reversion and trading profits strengthen the argument

The strongest evidence is not merely that overweighted stocks move more with the index today. It is that these same stocks then move *against* lagged index returns tomorrow, and that a simple strategy exploiting this reversal earns large abnormal profits.

Interpretation. If the contemporaneous relation were only about faster incorporation of information, one would not expect a strong cross-sectional reversal pattern or a profitable reversal strategy.

4.6 How the paper addresses nonsynchronous trading

A standard concern is that thinly traded stocks can appear to have lower contemporaneous beta and higher lagged beta because they incorporate information with a delay. The paper argues against this explanation because:

- the overweighted stocks in his data display *negative*, not positive, covariance with lagged index returns;
- the overweighting-beta relation is actually stronger among *high-volume* stocks;
- tick data show that on **99.536%** of ticker-days there is meaningful trading between 2:45 and 3:00 p.m., close to the market close.

Interpretation. Microstructure frictions are unlikely to be generating the core cross-sectional pattern.

4.7 What the paper can and cannot prove

- The paper does *not* directly observe investor flows into each stock.
- It also does not estimate the structural demand curve separately from the variance of index-trader shocks.
- What it does show is that a single exposure variable – index overweighting – predicts the whole sequence of facts: stronger index comovement, weaker nonindex comovement, stronger reversal, and profitable trading strategies.

Interpretation. The evidence is reduced-form but internally very coherent.

4.8 Relation to the literature

- Relative to Barberis, Shleifer, and Wurgler (2005), the paper replaces an event-study design with a cross-sectional design based on weight heterogeneity.
- Relative to the broader excess-comovement literature, it adds an explicit predictability test: if comovement is excessive, it should reverse.
- Relative to market microstructure explanations, it highlights demand commonality rather than only information incorporation or stale-price effects.

5 Tables and figures

5.1 Table 1: Sample composition and basic characteristics

Read:

- The main panel contains 298 stocks that enter the Nikkei 225 at some point during 1993–2003.
- The expanded sample adds 1458 never-in-index stocks.
- The table shows that index stocks are, on average, larger and more actively traded than non-index stocks.

Economic message. This table explains why the paper needs controls and expanded-sample checks: Nikkei and non-Nikkei stocks are not identical, so the cleanest comparisons come from variation *within* index members.

	Mean	SD	Min	Max
Main sample ($N = 298$)				
Length of time series (days)	2401	455	109	2609
Fraction of period in Nikkei (%)	75.50	32.16	4.18	100.00
Daily returns (%)	1.07	3.95	-14.24	25.73
Daily turnover (%)	1.00	5.57	0.02	68.47
Size (\$ million)	910, 229	1, 797, 490	19, 739	18, 884, 907
Ln (P/B)	0.57	0.42	-0.45	2.70
Leverage	0.94	0.14	0.00	1.00
Other stocks ($N = 1458$)				
Length of time series (days)	2154	780	0	2609
Daily returns (%)	0.43	0.10	-81.19	219.57
Daily turnover (%)	0.14	0.57	0.00	12.64
Size (\$ million)	145, 698	581, 337	1, 633	9, 912, 280
Ln (P/B)	0.28	0.57	-2.12	3.62
Leverage	0.82	0.25	0.00	1.00

Cross-sectional mean, standard deviation, and extreme values of time series averages of selected variables. The main sample includes 298 stocks that were members of the Nikkei 225 index for at least one day during the period from 1 September 1993 through 1 September 2003. This period includes a total of 2609 trading days. The second sample includes 1458 stocks in Japan that provided at least two years of returns data and that were never members of the index between September 1993 and September 2003. The length of the time series is the number of days for which each stock provides volume and price data. The fraction of the sample in Nikkei 225 is the percentage of the time that the stock was a member of the Nikkei 225 stock index. Daily return is the time series average of returns for each stock. Turnover is the average daily trading volume expressed as a percentage of total shares outstanding. Size is equal to the time series average of market capitalization, in millions of yen. The log price-to-book value is the natural log of the share price divided by the book value per share for the appropriate financial year end, adjusted for capital changes. Leverage is the ratio of long-term debt to long-term debt plus common (book) equity. All data are collected from Datastream. The history of index membership is constructed using the index membership changes given on the *Nikkei Asian* Shimbun webpage.

Table 2
Nikkei 225 index composition and weights

Sample	N	Price	Price Face Value \$	w^{N225}	$w^{N225} w^{TOPX}$	Overweighting $\log(1 + \frac{w^{N225}}{w^{TOPX}})$
Face value 50	280	1491	1491	0.58	6.00	1.62
Face value 200	4	320	332	0.18	0.83	0.46
Face value 5000	2	1,098,500	10,985	4.42	5.83	1.92
Face value 50,000	12	987,583	988	0.34	0.38	0.30
Full sample	298	48,586	5118	0.59	5.70	1.55

Relationship between prices and Nikkei 225 weights for the 298 stocks present in the Nikkei 225 stock index for at least one day between 1 September 1993 and 1 September 2003. The level of the Nikkei 225 stock index is given by

$$P_{Nikkei,t} = \frac{1}{D_t} \sum_{i=1}^{225} \frac{P_{i,t}}{P_{i,70}}$$

where D_t is the Nikkei 225 divisor, $P_{i,t}$ is the price of stock i on day t , and $P_{i,70}$ is the face value of stock i ranging from 50 to 50,000. The table reports average prices of stocks, when the price is taken from the first day of the sample in which a stock is present in the index. The next column reports mean stock price normalized by face value—this is the form in which prices enter the Nikkei 225 index calculation. The next column reports the average weight in the Nikkei index, the ratio of the face value adjusted price to the sum of the face value adjusted prices. The second-to-last column reports the average ratio of the Nikkei index weight to the weight in the market value-weighted TOPX index. The final column reports the average of the log of the ratio in the previous column. This is the measure of overweighting used in the article. Repeated averages are broken down by face value (\$5, 500, 5000, or 50,000).

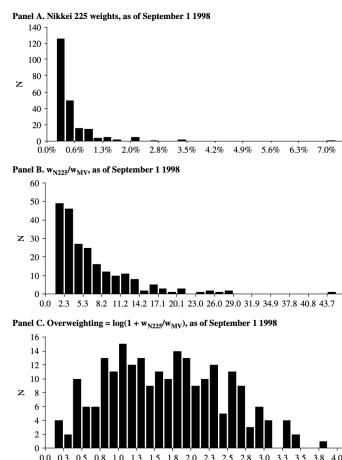


Figure 1
Nikkei 225 index weights and overweighting
Cross-sectional distribution of index weights of the stocks in the Nikkei 225 index on 1 September 1998, the midpoint of the panel. Panel A shows the distribution of raw index weights for the 225 stocks. Panel B shows the distribution of the ratio of index weights to the weight that each stock takes in the market value-weighted TOPX index. A value greater than 1.0 indicates overweighting relative to the value-weighted TOPX. Panel C shows the distribution of the overweighting measure, equal to the natural log of one plus the ratio of index weights to the weight that each stock takes in the value-weighted TOPX index.

5.2 Table 2 and Figure 1: How extreme Nikkei overweighting is

Read:

- Most constituents have face value 50, but a nontrivial minority have very different face values.
- The average ratio of Nikkei weight to value-weight benchmark weight is around **5.7**.
- Figure 1 shows that raw Nikkei weights are highly skewed, and the log-overweighting measure remains widely dispersed across constituents.

Economic message. The paper's key regressor is not a tiny perturbation. The Nikkei weighting rule creates economically large differences in exposure to index trading.

5.3 Figures 2 and 3 plus Table 3: Main cross-sectional comovement result

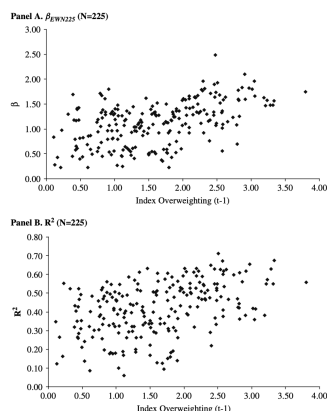


Figure 2
Comovement and Nikkei 225 index overweighting: Univariate regressions
Measures of comovement of stock returns with returns of other stocks in the Nikkei 225 are plotted against overweighting of the stock in the index. In Panel A, comovement is measured as the slope ($\beta_{EW225,t}$) from a time series regression of stock returns on the equal-weighted returns of stocks remaining in the Nikkei 225 throughout the sample period. In Panel B, comovement is measured as the R^2 from this regression. Index overweighting is defined as the natural log of one plus the ratio of a stock's weight in the Nikkei 225 to the weight of the stock in the value-weighted TOPIX index. Index overweighting is measured one day before the start of the sample of returns used to estimate comovement.

$$R_{it} = \alpha_i + \beta_{EW225,t} \cdot \text{Index Overweighting}_{t-1} + \epsilon_{it}$$

Both plots show comovement for the entire sample of Nikkei 225 stocks on 1 September 1998.

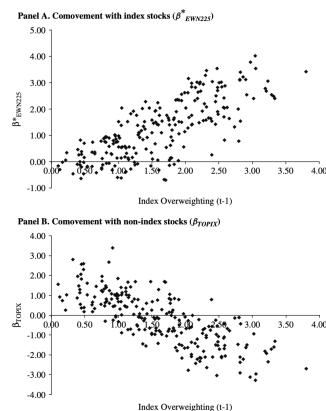


Figure 3
Comovement and Nikkei 225 index overweighting: Bivariate regressions
Measures of comovement of stock returns with returns of other stocks in the Nikkei 225 and with returns of stocks outside of the Nikkei 225 are plotted against overweighting of the stock in the index. The figure plots the slope coefficients ($\beta_{EW225,t}$ and $\beta_{TOPIX,t}$) from the regression

$$R_{it} = \alpha_i + \beta_{EW225,t} \cdot \text{Index Overweighting}_{t-1} + \beta_{TOPIX,t} \cdot \text{Index Overweighting}_{t-1} + \epsilon_{it}$$

against index overweighting, measured one day before the start of the sample of returns used to estimate comovement. Index overweighting is defined as the natural log of one plus the ratio of a stock's weight in the Nikkei 225 to the weight of the stock in the value-weighted TOPIX index. Regression slope parameters are estimated using 100 days of data. Panel A plots $\beta_{EW225,t}$ for every stock in the sample on 1 September 1998, the midpoint of the sample, against the index overweighting 100 days before. Panel B plots regression slope parameter $\beta_{TOPIX,t}$ against the index overweighting 100 days before.

Table 3 Nikkei 225 overweighting and comovement measures								
Y	a	[t-stat]	b	[t-stat]	c	[t-stat]	d	[t-stat]
β_{EW225}	0.719	[24.38]	0.218	[10.17]				
β_{TOPIX}	2.401	[9.92]	0.048	[1.84]	0.005	[1.22]	-0.113	[-6.68]
R^2	0.209	[12.98]	0.065	[4.31]				
R^2	0.340	[4.50]	0.053	[6.44]	0.005	[2.14]	-0.004	[-0.84]

Average parameter estimates and Fama and MacBeth (1973) t -statistics from rolling cross-sectional regressions of comovement measures on index overweighting at the start of the period

$$\beta_{EW225,t} = \alpha_i + \beta_1 \cdot \text{Index Overweighting}_{t-1} + \beta_2 \cdot \ln(P/B)_{t-1} + \beta_3 \cdot \text{Size}_{t-1} + \epsilon_{it}$$

$$R^2_{it} = \alpha_i + \beta_1 \cdot \text{Index Overweighting}_{t-1} + \beta_2 \cdot \ln(P/B)_{t-1} + \beta_3 \cdot \text{Size}_{t-1} + \epsilon_{it}$$

β_{EW225} denotes the slope parameter from a regression of daily stock returns on the equal-weighted returns of stocks that were in the Nikkei 225 index for the entire sample period, and R^2 denotes the R^2 from this univariate regression. First-stage regression estimates for period t are computed using returns from the interval $[t-99, t]$ and are estimated for each stock for each 100-day interval. The table shows average coefficients from second-stage cross-sectional regressions of $\beta_{EW225,t}$ and R^2_{it} on lagged independent variables. These include Nikkei 225 overweighting at the start of the period (OW), defined as the natural log of one plus the ratio of index weight to the weight the stock would have taken in a value-weight index, the natural log price-to-book ratio (P/B), and the natural log of market value ($Size$). All independent variables are measured at $t-100$, with the exception of the log price-to-book ratio, which is measured in December of the previous year. Overweighting is defined to be zero for firms that were not in the index at the start of the period. The table also indicates the average number of firms in each cross-section (N), as well as the time series average of R^2 from the cross-sectional regressions.

Read:

- Figure 2 shows a strong positive relation between overweighting and univariate Nikkei beta, and also between overweighting and the R^2 from the univariate regression.
- In Table 3, the average slope of beta on overweighting is about **0.218** with a Fama-MacBeth t -stat around **10.17**.
- The corresponding cross-sectional R^2 is about **0.21**, so overweighting alone explains more than one-fifth of the cross-sectional variation in daily comovement among index stocks.
- With size and price-to-book controls, the coefficient on overweighting falls but remains positive and significant (about **0.048**).

Economic message. The central fact of the paper is already visible here: stocks that index investors buy in larger proportions move more with the index.

5.4 Table 4: Conditional comovement with Nikkei versus nonindex stocks

Read:

- In Panel A, overweighting strongly raises conditional comovement with the equal-weighted Nikkei return.
- In Panel B, overweighting strongly *reduces* conditional comovement with the TOPIX return.
- The sign pattern survives controls for size and price-to-book.

Economic message. This is the paper’s cleanest reduced-form fingerprint of common demand. Overweighted stocks comove more with the index basket and less with the rest of the market.

Table 4
Nikkei 225 overweighting and conditional comovement

Y	a	[t-stat]	b	[t-stat]	c	[t-stat]	d	[t-stat]	N	R ²
Panel A: Daily conditional comovement with Nikkei 225 stocks										
β_{EWN225}^*	-0.159	[-2.30]	0.800	[16.28]					222	0.390
$\beta_{EWN225}^{\dagger}$	6.748	[17.39]	0.136	[2.24]	-0.062	[-2.97]	-0.465	[-17.55]	222	0.614
Panel B: Daily conditional comovement with stocks outside of Nikkei 225										
β_{TOPIX}^*	1.178	[13.53]	-0.780	[-13.06]					222	0.330
$\beta_{TOPIX}^{\dagger}$	-5.661	[-12.83]	-0.126	[-1.81]	0.077	[2.54]	0.459	[15.29]	222	0.530

In the first stage, I jointly estimate the conditional comovement of a stock's return with the equal-weighted return of the stocks present in the Nikkei 225 index and the return on the TOPIX value-weighted index.

$$R_{it} = a_i + \beta_{EWN225, it}^* R_{EWN225, it} + \beta_{TOPIX, it}^* R_{TOPIX, it} + \epsilon_{it}$$

The table reports average parameter estimates and Fama and MacBeth (1973) *t*-statistics from second stage rolling cross-sectional regressions of conditional comovement on index overweighting at the start of the period as well as a set of control variables

$$\beta_{EWN225, it}^* = a_i + b_i \cdot OW_{it-1} + c_i \cdot \ln(P/B)_{it-1} + d_i \cdot Size_{it-1} + u_{it}$$

$$\beta_{TOPIX, it}^* = a_i + b_i \cdot OW_{it-1} + c_i \cdot \ln(P/B)_{it-1} + d_i \cdot Size_{it-1} + u_{it}$$

where β_{EWN225} and β_{TOPIX} are the conditional comovement estimates from the first-stage regressions. The independent variables in the second-stage regressions include Nikkei 225 overweighting at the start of the period (OW), defined as the log of one plus the ratio of index weight to the weight the stock would have taken in a value weighted index, the price-to-book ratio (P/B), and the log of market value ($Size$). All independent variables are measured at $t-100$, except for the price-to-book ratio, measured in December of the previous year. For each cross-section, the sample includes all firms present in the Nikkei index at the beginning and end of the estimation period. The table indicates the average number of firms in each cross-section (N), as well as the time series average of R^2 from the cross-sectional regressions.

Table 5
Nikkei 225 index overweighting and comovement: Expanded sample

Dependent variable (β)	First-stage regression	β Measures comovement with:	a	[t-stat]	b	[t-stat]	c	[t-stat]	d	[t-stat]	e	[t-stat]	N	R ²
Panel A: Comovement with Nikkei 225 members (univariate)														
β_{EWN225}^*	Univariate	Nikkei stocks	0.577	[27.12]	0.296	[22.05]							1421	0.229
$\beta_{EWN225}^{\dagger}$	Univariate	Nikkei stocks	0.372	[26.75]	0.218	[10.17]	0.147	[5.81]					1421	0.237
$\beta_{EWN225}^{\ddagger}$	Univariate	Nikkei stocks	0.656	[5.61]	0.202	[7.90]	0.186	[4.02]	0.030	[2.77]	-0.010	[-1.00]	1421	0.266
Panel B: Conditional comovement with Nikkei 225 members (multivariate)														
β_{EWN225}^*	Multivariate	Nikkei stocks	0.106	[2.54]	0.664	[20.50]							1416	0.295
$\beta_{EWN225}^{\dagger}$	Multivariate	Nikkei stocks	0.115	[2.78]	0.800	[16.28]	-0.275	[-5.06]					1416	0.301
$\beta_{EWN225}^{\ddagger}$	Multivariate	Nikkei stocks	1.065	[5.81]	0.666	[9.52]	0.113	[1.07]	-0.050	[-1.80]	-0.091	[-3.92]	1416	0.368
Panel C: Conditional comovement with non-Nikkei 225 stocks (multivariate)														
β_{TOPIX}^*	Multivariate	Non-Nikkei stocks	0.634	[15.25]	-0.496	[-14.65]							1416	0.141
$\beta_{TOPIX}^{\dagger}$	Multivariate	Non-Nikkei stocks	0.614	[14.71]	-0.780	[-13.06]	0.564	[7.14]					1416	0.158
$\beta_{TOPIX}^{\ddagger}$	Multivariate	Non-Nikkei stocks	-0.524	[-2.00]	-0.623	[-8.25]	0.099	[0.94]	0.082	[3.83]	0.106	[4.83]	1416	0.238

Average parameter estimates and Fama and MacBeth (1973) *t*-statistics from rolling nonoverlapping cross-sectional regressions of comovement of daily stock returns on index overweighting at the start of the period, a dummy variable for index membership, and controls

$$\beta_{EWN225, it}^* = a_i + b_i \cdot OW_{it-1} + c_i \cdot \ln(P/B)_{it-1} + d_i \cdot \ln(P/B)_{it-1} + e_i \cdot Size_{it-1} + u_{it}$$

β denotes the comovement measure, estimated in one of three ways. In Panel A, β is the slope parameter from a univariate regression of stock returns on the equal-weighted return of stocks that were in the Nikkei 225 for the entire sample period. In Panel B, β is the coefficient on equal weighted Nikkei return, from a multivariate regression of stock returns on the equal-weighted return Nikkei 225 stocks and the broader value-weighted TOPIX index. In Panel C, β is the coefficient on the TOPIX return, from the same multivariate regression. Thus in Panel C, β measures comovement with stocks outside of the Nikkei stock index. In all cases, β is estimated using returns from the interval $t-99$ to t , and the independent variables are measured at $t-100$, except for the price-to-book ratio, which is measured in December of the previous year. Each cross-section contains all the firms in Japan that contain complete data and for which β can be estimated. The independent variables include Nikkei 225 overweighting at the start of the period (OW), defined as the natural log of one plus the ratio of index weight to the weight the stock would have taken in a value-weighted index, a dummy variable indicating whether the firm is present in the Nikkei index (I_{Nikkei}), the natural log of the price-to-book ratio (P/B), and the log of market value ($Size$). Overweighting is zero for firms that were not in the index at the start of the period. The last two columns indicate the average number of firms in each cross-section (N), as well as the time series average of R^2 from the cross-sectional regressions.

5.5 Table 5: Weighting matters even after controlling for membership

Read:

- In the expanded sample, the coefficient of univariate Nikkei beta on overweighting is about **0.296** without other controls.
- After adding an index-membership dummy, the slope remains about **0.218**; with additional controls it is still about **0.202**.
- The index-membership dummy itself becomes weak or even takes the wrong sign once overweighting is included.

Economic message. Being in the index matters far less than *how heavily weighted* a stock is within the index. This is exactly the advantage of the cross-sectional design.

5.6 Table 6: Excess comovement is weaker at longer horizons

Read:

- The overweighting-comovement relation remains positive using weekly and monthly returns.
- But the statistical strength of the relation is lower than in the daily data, especially in the all-stocks sample.
- The paper interprets this as evidence that daily excess comovement is gradually corrected over time.

Economic message. Demand-driven comovement is mainly a short-horizon phenomenon. By the time one looks at longer-horizon returns, part of the distortion has already reversed.

Table 6 Comovement measured at longer horizons					
Y	a	[t-stat]	b	[t-stat]	N
Daily return comovement					
Nikkei only	0.719	[24.38]	0.218	[10.17]	0.21
All stocks	0.577	[27.12]	0.296	[22.05]	0.23
Weekly return comovement					
Nikkei only	0.773	[11.58]	0.180	[3.98]	0.15
All stocks	0.709	[17.92]	0.217	[8.50]	0.10
Monthly return comovement					
Nikkei only	0.694	[14.86]	0.237	[6.90]	0.19
All stocks	0.897	[10.94]	0.135	[4.85]	0.03

Average parameter estimates and Fama and MacBeth (1973) *t*-statistics from rolling nonoverlapping cross-sectional regressions of comovement of daily stock returns on index overweighting at the start of the period

$$\beta_{EWN225, it} = a_i + b_i \cdot OW_{it-1} + u_{it}$$

β_{EWN225} denotes the slope parameter from regressions of daily, weekly, and monthly stock returns on the equal-weighted return of stocks that were in the Nikkei 225 index for the entire sample period. First-stage regression estimates for period *t* are computed using returns from the interval [*t* - 90, *t*] and are estimated for each stock for each 100-day interval. The table shows average coefficients from second-stage cross-sectional regressions of β_{EWN225} on overweighting. Overweighting is the natural log of one plus the ratio of index weight to the weight the stock would have taken in a value-weighted index. The table also indicates the average number of firms in each cross-section (N), as well as the time series average of R^2 from the cross-sectional regressions.

Table 7
Reversion of excess comovement

Panel A: Sample averages			
	$\beta_{EWN225, it}$	[t-stat]	$\beta_{EWN225, it-1}$
Full sample	1.05	[105.12]	-0.008
OW < Median(OW)	0.92	[74.63]	0.05
OW ≥ Median(OW)	1.18	[54.76]	-0.066
Panel B: Cross-sectional relationships			
	a	[t-stat]	b
$\beta_{EWN225, it}$	0.72	[22.88]	0.22
$\beta_{EWN225, it-1}$	0.13	[6.39]	-0.09

In the first stage, I estimate the contemporaneous comovement of the stock return with the equal-weighted return of the stocks in the Nikkei 225 index and the tendency to revert this comovement the next day

$$R_{it} = a_i + \beta_{EWN225, it} \cdot R_{EWN225, it} + \beta_{EWN225, it-1} \cdot R_{EWN225, it-1} + \epsilon_{it}$$

These regressions are estimated every 100 days for every stock in the sample. The first line of Panel A reports the time series average of the cross-sectional mean of contemporaneous and lagged betas from this regression. The second and third lines of Panel A report these averages sorted by the overweighting of each stock in the Nikkei index. Overweighting is defined as the natural log of one plus the ratio of index weight to the weight the stock would have taken in a value-weighted index. Stocks are sorted in each cross-section as being above or below the median overweighting. Panel B reports average coefficients from cross-sectional regressions of these beta coefficients on overweighting at the start of the period.

$$\beta_{EWN225, it} = a_i + b_i \cdot OW_{it-1} + u_{it}$$

$$\beta_{EWN225, it-1} = a_i + b_i \cdot OW_{it-1} + u_{it}$$

These regressions are estimated on each of the 100-day nonoverlapping cross-sections between 1993 and 2003. Fama-MacBeth *t*-statistics are reported in brackets.

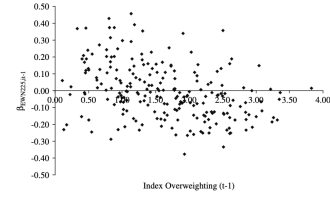


Figure 4
Reversion of excess comovement
A measure of the reversion of excess comovement with the Nikkei 225 stock index is plotted against lagged overweighting. The figure plots the slope coefficients on the lagged index return, $\beta_{EWN225, it-1}$, from the multivariate regression

$$R_{it} = a_i + \beta_{EWN225, it} \cdot R_{EWN225, it} + \beta_{EWN225, it-1} \cdot R_{EWN225, it-1} + \epsilon_{it}$$

against index overweighting, measured one day before the start of the sample used to estimate coefficients. Index overweighting is defined as the log of one plus the ratio of a stock's weight in the Nikkei 225 to the weight of the stock in the value-weighted TOPIX index. Regression slope parameters are estimated using 100 days of data ending at the midpoint of the panel, on 1 September 1996.

5.7 Table 7 and Figure 4: Reversion of excess comovement

Read:

- For the full sample of index stocks, contemporaneous beta on the equal-weighted Nikkei return is about **1.05**, while the average lagged-index beta is slightly negative (about **-0.008**).
- Stocks below the median overweighting have contemporaneous beta about **0.92** and lagged beta about **+0.05**.
- Stocks above the median overweighting have contemporaneous beta about **1.18** and lagged beta about **-0.066**.
- In the cross-section, overweighting loads positively on contemporaneous beta (about **0.22**) and negatively on lagged beta (about **-0.09**).
- Figure 4 shows that the most overweighted stocks can reverse as much as about **30%** of the prior index move the next day.

Economic message. The stocks that comove most because of index demand are also the stocks whose movements are least permanent.

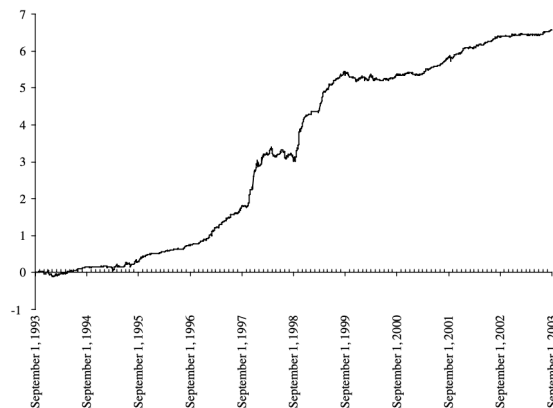


Figure 5
Cumulative profits from reversion of excess comovement trading strategy
 Cumulative profits from a trading strategy that goes short overweighted Nikkei 225 stocks following index increases (hedged by offsetting long positions in underweighted stocks) and goes long overweighted Nikkei 225 stocks following index declines (hedged by offsetting short positions in underweighted stocks). The weights of the zero investment portfolio are given by

$$w_{it} = - \left[OW_{it-1} - \frac{1}{N} \sum_N OW_{jt-1} \right] \cdot R_{EWN225,t-1}$$

where OW_{it-1} denotes index overweighting at the end of the previous day, and $R_{EWN225,t-1}$ denotes the lagged return on the equal-weighted portfolio of stocks in the Nikkei 225 stock index. Overweighting is defined as the log of one plus the ratio of Nikkei 225 index weight to the weight the stock has in the value weighted TOPIX index. Panel A shows cumulative profits from this strategy, in yen. The annualized Sharpe ratio corresponding to the figure is 2.56.

5.8 Figure 5: Profits from trading the reversal

Read:

- A zero-investment strategy that shorts overweighted stocks after index rises and buys them after index falls produces steadily rising cumulative profits.
- The paper reports an annualized Sharpe ratio above **2.3** for this reversal strategy.
- Profit spikes are especially visible around major liquidity events, such as the 1997–1998 crisis period.

Economic message. The paper does not stop at documenting unusual correlations. It shows that these correlations imply economically meaningful mispricing.

6 Short summary

- The Nikkei 225 is price weighted rather than value weighted, so some constituents are heavily overweighted relative to their economic size.
- If common index-trader demand matters, those overweighted stocks should comove more with the index and less with nonindex stocks.
- That is exactly what the data show in repeated 100-day cross-sections from 1993 to 2003.
- The relation survives controls for size, price-to-book, leverage, and index membership.

- Overweighted stocks also reverse more strongly after index moves, which is difficult to reconcile with a pure-fundamentals story.
- A simple reversal strategy based on overweighting earns large abnormal profits.
- The evidence therefore supports a demand-based explanation of excess comovement, rather than one based only on common fundamentals or information diffusion.

7 Conclusion

7.1 Core conclusion

The paper shows that cross-sectional variation in Nikkei 225 weights predicts the cross-section of stock-return comovement. Stocks that are more overweighted in the index comove more with other Nikkei stocks, comove less with nonindex stocks, and later reverse more strongly.

7.2 Economic intuition

The simplest interpretation is that index traders buy and sell Nikkei stocks together, but not in equal proportion. Because arbitrage is limited, these demand shocks move the prices of heavily weighted stocks more strongly. The resulting correlation structure is therefore partly a product of common trading behavior rather than purely common fundamentals.

7.3 Takeaway

The paper's deeper message is that index construction is not innocuous. Arbitrary weight rules can affect how prices move together and can create predictable reversals. For empirical finance, this means that observed comovement is not always a clean statistic for shared cash-flow or discount-rate news; some of it is generated mechanically by the trading habits induced by benchmark design.